Extension: Special right triangles

Core Concepts: Extension: Special right triangles

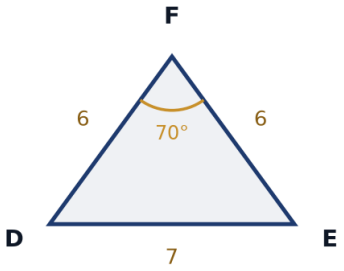
1 In $\triangle ABC$ studying extension: special right triangles, $m\angle A = 48^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 87°

b. 97°

c. 79°

d. 132°



2 A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm

c. 21 cm

d. 28 cm

3 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 1$?

a. 4

b. 11

c. 9

d. 14

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying extension: special right triangles, $m\angle A = 48^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 80°

b. 90°

c. 72°

d. 132°

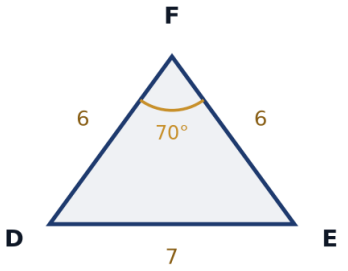
5 A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length?

a. 18 cm

b. 21 cm

c. 28 cm

d. 63 cm



6 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 2$?

a. 9

b. 16

c. 14

d. 19

Graphical & Geometric Interpretation

7 In $\triangle ABC$ studying extension: special right triangles, $m\angle A = 48^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 73°

b. 83°

c. 65°

d. 132°

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8 A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length?

a. 24 cm

b. 28 cm

c. 35 cm

d. 112 cm

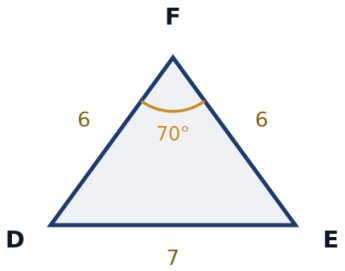
9 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$?

a. 14

b. 21

c. 19

d. 24



Exam-Style Problem Solving & Modeling

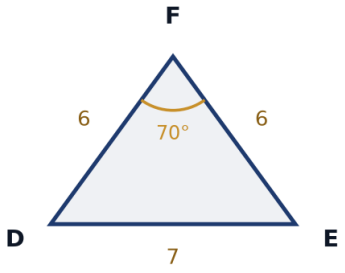
10 In $\triangle ABC$ studying extension: special right triangles, $m\angle A = 48^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 66°

b. 76°

c. 58°

d. 132°



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11 A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length?

a. 30 cm

b. 35 cm

c. 42 cm

d. 175 cm

12 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$?

a. 19

b. 26

c. 24

d. 29

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